Anatomically Accurate Head Model for Traumatic Brain Injury (TBI) Research

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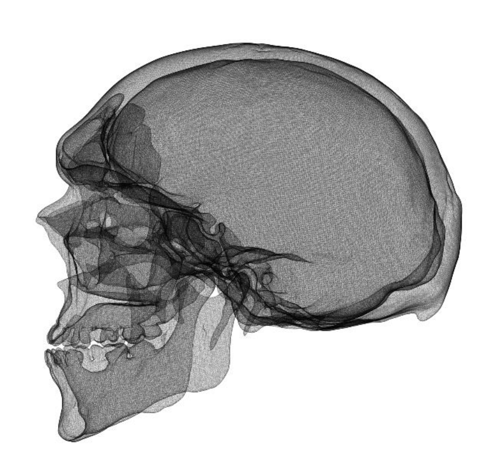
BACKGROUND

Problem	Hypothesis	Current Limitations	Approach
>50 million TBI cases worldwide annually but **still no cure**	**Increasing intracranial pressure (ICP) via the Valsalva maneuver (VM) may have neuroprotective properties**	Existing head models **cannot manipulate ICP**	**Design and assemble multi-layer physical head models with ICP control**

HEAD MODEL DESIGN AND FABRICATION

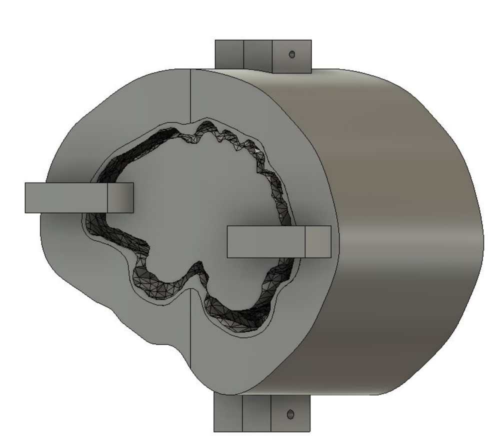
- **Skull**
  - Modified from CT scan-derived NIH3D model on Fusion360
  - 3D printed in nylon



Original Skull

3D Printed Skull

- **Brain**
  - Cast using polyacrylamide (PAA) hydrogel
  - Tuned for gray and white matter



Custom Mold

Cast Brain

- **Scalp + Dura Mater**
  - Scalp pads made from Vytaflex40
  - Dura mater 3D printed in Elastico

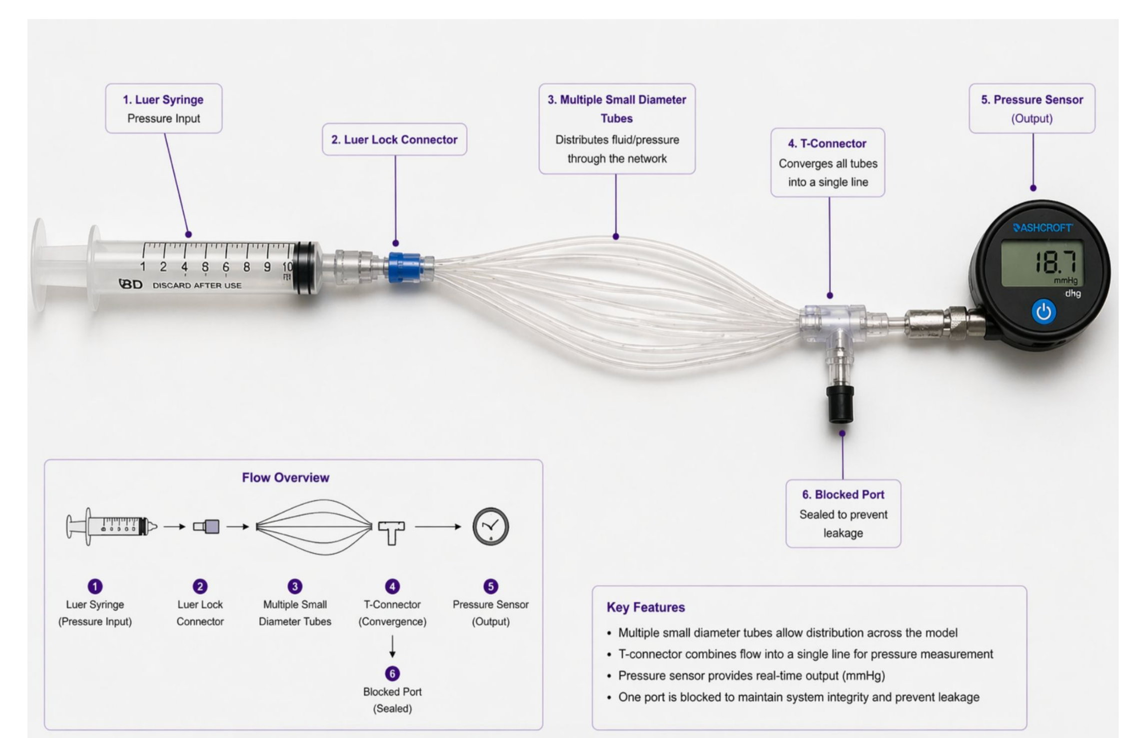
Scalp

Dura Mater

All materials chosen have similar mechanical properties to their real human counterparts.

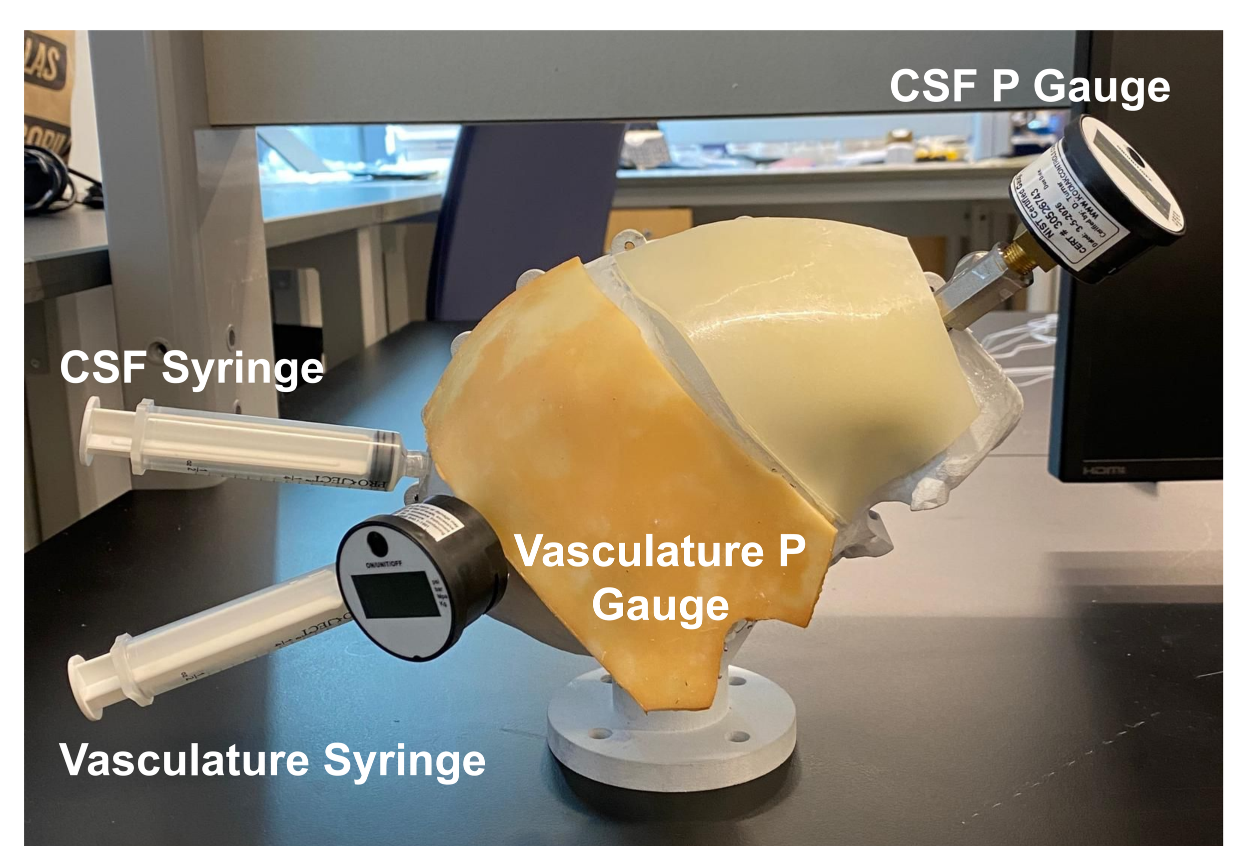
PRESSURE SYSTEM

- **Vasculature Network:** silicone tubing



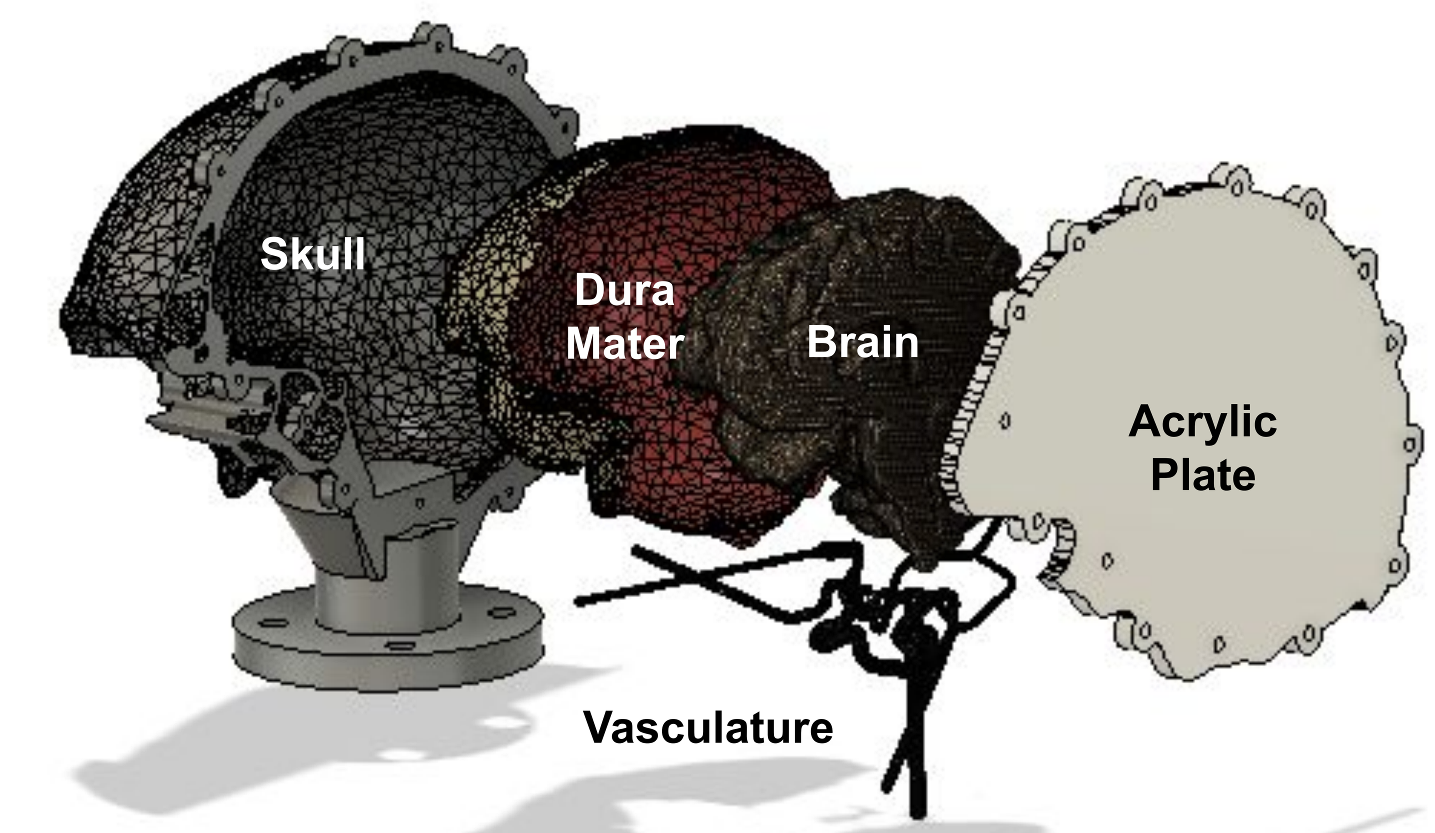
AI-Generated Image

- **CSF Pressure:** Inject water through hole to increase CSF pressure
- **Vasculature Pressure:** Inject water into tubing to increase vasculature pressure



MODEL DEVELOPMENT PROGRESS

- ✔ Designed and fabricated ½ skull model
- ✔ Developed PAA brain surrogate with embedded vasculature
- ✔ Fabricated dura + scalp
- ✔ Integrated a pressure system for VM simulation
- ❏ Full assembly



FUTURE WORK

- Perform four blunt impact experiments to test the individual and combined effects of CSF and vasculature pressure on skull and brain deformation

	CSF Pressure	Vasculature Pressure
Control	✗	✗
Exp. 1	✓	✗
Exp. 2	✗	✓
Exp. 3	✓	✓